ARTICLE

PRISM: a modular system for glioma progression surveillance on MRI

[PENDING: first author's full name][a] and Jessica Zhang[b]

[a]Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA; [b]Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA

ARTICLE HISTORY

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ABSTRACT

[PROPOSED, awaiting approval] Most glioma surveillance datasets lack treatment dates, centres document progression differently, cohorts are small and the hardest scans are few. We built PRISM, a modular system that detects the scan at which progression is documented through separate image and treatment-timeline pathways, attaches a confidence to each scan and writes a report with a local language model. On MU-Glioma-Post the timeline pathway reached an AUC of 0.82, above the image pathway. Fitted once and applied unchanged, the models reached 0.77 on LUMIERE; on UCSF-ALPTDG, single-scan measurements reached 0.80 and the image pathway with change from the previous scan 0.73, against 0.86 when trained there. An audit showed the scores rested mainly on the treatment timeline at MU-Glioma-Post and on enhancing tumour volume at UCSF-ALPTDG. Conformal sets for the combined score met their pre-specified coverage at all three centres, and after one revision every report passed a deterministic check against its record.

KEYWORDS

glioma surveillance; external validation; clinical decision support

1. Introduction

After surgery and chemoradiotherapy, patients with glioma are followed with MRI every few months, and at each scan the team must decide whether the tumour has progressed. The decision is hardest in the months after radiotherapy, when treatment can increase contrast enhancement without tumour growth. This pseudoprogression is most common in the first weeks after chemoradiotherapy and more frequent when the MGMT promoter is methylated (Taal et al. 2008; Brandes et al. 2008). In routine care, progression is therefore documented from the images together with symptoms and the course of treatment, and the date on which it is recorded depends on when the patient was scanned, so a model trained on such records can learn from the schedule as well as from the images.

Models for this task meet five challenges that come from the data rather than from any one method. First, most public surveillance datasets hold images but few treatment dates, so a model that needs the treatment timeline cannot be applied to them. Second, centres document progres-

CONTACT [PENDING: first author. Email: ...; ORCID: ...]

sion in different ways, from dated clinical decisions to expert ratings of each scan, so a model can score well at two centres for different reasons. Third, cohorts are small, and at this size the choice of one test split can move an AUC by more than a change of model (An et al. 2021; Varoquaux 2018). Fourth, the situations in which the decision is hardest, early after radiotherapy or when enhancement grows, hold few cases in any one cohort, so accuracy there is rarely measured. Fifth, tumour volumes from automatic segmentation are noisy, so the clinician needs to see what was measured; and patient records often cannot leave the hospital, so any report must be written inside it. External tests address the second and third challenges. In a recent systematic review of automated response assessment, 3 of 20 studies used external test data (Shi et al. 2025), and deep learning models in radiology usually lose some accuracy on external data (Yu et al. 2022).

We asked whether a system could give the clinician, at each post-treatment scan, the image evidence and the treatment-timeline evidence separately, transfer to other centres without refitting, and say how far each can be trusted there. We also wanted it to explain each score in a short report written inside the hospital.

PRISM is a modular system built for this. We developed it on MU-Glioma-Post (MU) and tested it on LUMIERE and UCSF-ALPTDG (UCSF) with models fitted once and external tests planned in advance. It scores each scan through two pathways, one from clinical data and the treatment timeline and one from measurements of the segmented tumour, and shows a combined score beside them. It attaches a confidence made of a conformal prediction set, a statement of whether the pathways agree and a reliability card, and writes a report with a local language model whose every sentence is checked against the scan's record. On MU both pathways detected the scan at which progression was documented; the models fitted there reached an AUC of 0.77 at LUMIERE, and the image measurements transferred to UCSF. An audit of the scores showed that they rested mainly on the patient's position in treatment at MU and on the enhancing tumour volume at UCSF. The conformal sets for the combined score met their pre-specified coverage at all three centres, and after one revision of the prompt, following a failure at one cohort, the reports were faithful to their records.

2. Methods

2.1. PRISM

PRISM handles each scan in five modules (Figure 1). The timeline pathway and the image pathway keep the two kinds of evidence apart, so that the image pathway can run where no treatment dates exist and the audit can show which evidence a score used. Segmentation masks supply the measurements and the images the clinician sees. The confidence addresses the rare situations, and the report presents all of it without the record leaving the hospital. The third challenge, small cohorts, is handled by how PRISM is evaluated (Section 2.3).

The timeline pathway scores a scan from clinical data and the patient's position in treatment. On MU its inputs are age, sex, IDH mutation and MGMT promoter methylation, which alone form the clinical-only reference model, together with the days since diagnosis and since surgery, the start, end and dose of radiotherapy, chemotherapy and the visit index, each taken as of the scan day. Where a cohort records no treatment dates, PRISM uses a reduced timeline that keeps the clinical inputs, the days since surgery and a flag for the first weeks after radiotherapy. At LUMIERE

the flag is the rater's note that the scan was less than three months after radiotherapy; at UCSF it is approximated from the start of radiotherapy. Both pathways are ridge logistic regressions, with every fitted step inside the training data.

The image pathway scores a scan from simple measurements of the segmented tumour. On MU it uses 27 inputs: the log volumes of six tumour compartments, including enhancing tumour, necrosis, the surrounding FLAIR hyperintensity and the resection cavity, with shape and relative intensity features; the log ratio of each volume to the previous scan; and the ratio of the enhancing and total lesion volumes to the smallest volume in the patient's history. It holds no dates, intervals or treatment flags. A single-scan version keeps only the 19 measurements of the current scan. At LUMIERE, whose masks have three labels, the pathway uses three volumes and their change from the previous scan, and at UCSF the history is the one earlier scan in the dataset. We defined these inputs after finding that the planned imaging model contained inputs that encode the scan schedule (Section 3.3), so the image pathway on MU is exploratory; its use at UCSF and in the system tests was planned. The combined score is one ridge model fitted on both sets of inputs. At LUMIERE and UCSF its inputs are the reduced timeline with harmonised volumes, their change and the interval since the previous scan, so there it holds one schedule input, that interval.

The measurements come from the segmentation masks supplied with each cohort: the automatic masks released with MU (Mahmoud et al. 2025), the DeepBraTumIA masks of LUMIERE (Suter et al. 2022) and the masks provided with UCSF (Fields et al. 2024). PRISM renders the report's figure from the same masks by code, and the language model receives only the record, not the image.

Each scan's confidence has three parts, a conformal set, a pathway-agreement statement and a reliability card. The conformal set is a patient-level split conformal prediction set at the 80% level (Angelopoulos and Bates 2023). It holds one outcome when the score is clear and both outcomes when it is not, and it should contain the true outcome for at least 80% of scans. The set comes from the combined model refitted on 70% of the MU patients and calibrated on the remaining 30%; on MU this split is made inside each training fold, and for the external cohorts one MU-calibrated set was applied unchanged. Because of the refit, the set's model ranks scans slightly differently from the displayed combined score (Supplementary Section [S24, system results, to be added]). The two pathways agree when both scores lie on the same side of that pathway's median score over the cohort's evaluation scans; the rule uses scores only, never labels. The reliability card looks up the scan's situation, its phase of care crossed with the change in enhancing tumour since the previous scan, in a table of results measured in each cohort, in the spirit of a model card (Mitchell et al. 2019). It states the situation's tier: tested, with AUCs and intervals, when the situation holds at least 10 progression and 10 non-progression scans; few cases, with counts only, when it holds fewer; and not tested when it holds none. The agreement rule and the card were defined after the cohort results had been seen and are exploratory.

A local language model writes the report from the scan's record, and a deterministic checker tests every sentence against that record. The model (Qwen3.8-27B) runs offline at temperature 0 and receives only the record; the checker rejects any number, flag or confidence statement that is not in the record, any omission of required content, and any wording that offers a diagnosis, advice or a probability (Supplementary Section S19).

2.2. Cohorts and the task

PRISM was developed on MU and tested on LUMIERE and UCSF with models fitted once on MU and applied unchanged, which we call locked (Table 1). MU records the dates of surgery, radiotherapy and chemotherapy, and progression as a documented date (Mahmoud et al. 2025). LUMIERE records no treatment dates; each scan has an expert rating of response, and the rater notes whether the scan falls within three months of radiotherapy (Suter et al. 2022). UCSF holds one pair of consecutive post-treatment scans per patient, with the date of progression and the start of radiotherapy (Fields et al. 2024). Because the first scan of each pair has no earlier scan, we evaluated the second scan only. UCSF was obtained under a data use agreement. The primary population in every cohort is grade 4 glioma.

The task is detection at the scan: is this the scan at which progression is documented? On MU a positive is the first scan on or after the documented progression date, and a negative is a scan followed by at least 120 days without progression. At LUMIERE a positive is the first scan rated as progressive disease, and a negative is an earlier scan rated stable or responding. At UCSF a positive is a second scan with progression dated within 30 days of it and nearer to it than to the first scan; a negative is a second scan with progression at least 120 days later, or with at least 120 days of follow-up without progression. On MU the intended use is scans from the start of radiotherapy onwards, and every MU result below is for this population; LUMIERE and UCSF are evaluated on all scans in their detection sets. We divide the intended use into phases of care: during radiotherapy, early after radiotherapy (up to 84 days, 12 weeks, after its end) and later. Predicting progression within 120 days of a scan, the task with which this work began, is reported as a secondary task.

2.3. Evaluation

On MU, every model was evaluated by repeated patient-grouped cross-validation. We used five repeats of five folds, with all scans of a patient in the same fold, and report the mean over repeats of the pooled held-out AUC with a 95% confidence interval (CI) from 1,000 patient-level bootstrap draws. In no analysis does a patient appear in more than one of fitting, calibration and testing. We chose this design because in cohorts of this size one split can move an AUC by more than a change of model. We saw this in our first evaluation, in which a radiomics model reached an AUC of 0.80 on one test split and averaged 0.72 when re-selected and re-tested over 500 splits of the same design (Supplementary Section S8 and Figure S2).

For the external tests, each model was fitted once on MU and applied once, under plans committed before any external label was read. The locked models were fitted on all grade 4 detection scans of MU, progression and non-progression scans from every phase of care; the UCSF plan named all MU detection scans, and fitting on all gliomas gave the same results. External AUCs have 95% CIs from 1,000 patient-level bootstrap draws. As a reference, we also fitted each UCSF model on UCSF itself by repeated cross-validation. Table 2 lists the questions fixed in advance and their answers.

Analyses that were not planned are labelled exploratory wherever they appear. They are the definition of the image pathway on MU, the three-step comparison of image models, the clinical-only, image-only and timeline-only models at LUMIERE, the single-scan comparison with models trained at UCSF, the audits of what each score rests on, the agreement rule and the reliability card, a stricter flag rule, and the second evaluation of the reports, which followed a failure seen in the first. Comparisons within a phase of care or within an enhancement category are diagnostics; they

test what a score rests on and set no decision rule.

3. Results

3.1. Both pathways detect the progression scan on MU

In the intended use on MU (146 scans; Table 1), the timeline pathway detected the scan at which progression was documented with an AUC of 0.82 (95% CI 0.75–0.90), and the image pathway with 0.71 (0.63–0.78). Clinical data alone, without the timeline, scored 0.53, close to chance. The combined score also reached 0.82; on MU the images did not raise the timeline's AUC, and the planned imaging model fell short of the pre-specified gain of 0.03 (Table 2). Over all scans of all gliomas the planned model did gain 0.04 (0.01–0.08), through image inputs that encode the scan schedule (Section 3.3). What the images contribute on their own is examined within phases of care in Section 3.3. Predicting progression within 120 days was harder; PRISM's model reached 0.60 (0.52–0.69) in the intended use (Supplementary Section S23). A timeline pathway needs treatment dates, which the two external cohorts largely lack, so we next tested how each pathway transfers to them.

3.2. The models transfer to two external centres without refitting

Fitted once on MU and applied unchanged, the combined model reached 0.77 (0.68–0.86) on LUMIERE (Figure 2), about as high as in cross-validation on MU, and met its planned transfer rule (Table 2). At LUMIERE each pathway added to the other; images alone scored 0.65 and the reduced timeline alone 0.69 (both exploratory), below the combined score. Clinical data alone stayed close to chance in all three cohorts (0.55 at LUMIERE, exploratory, and 0.50 at UCSF).

At UCSF, both planned image models transferred. Measurements of the single scan reached 0.80 (0.69–0.89), and the image pathway, which adds change from the previous scan, reached 0.73 (0.60–0.84), meeting its planned transfer rule (Table 2). Training at UCSF raised the image pathway to 0.86 (difference 0.14, 0.03–0.26), while the single-scan model scored about the same when trained there (0.83, exploratory). The change inputs therefore transferred less well than the measurements of the scan itself, which is consistent with the change between the two UCSF scans spanning a much narrower range than at MU. The reduced timeline transferred least, at 0.56, because UCSF records only the start of radiotherapy. These results show that PRISM's measurements transfer to a new centre, and that a model trained at the centre can do better. Because similar AUCs at different centres can rest on different evidence, we audited what each score used.

3.3. What each score rests on depends on the cohort

The audit (exploratory) shows that at MU the scores rest mainly on where the patient is in treatment, and at UCSF on how much the tumour enhances (Figure 3). At MU the scan schedule alone separates most of the two classes; most progression scans came more than 12 weeks after radiotherapy and most non-progression scans came within those 12 weeks (Table 1). Across phases of care, adding the timeline to clinical data and images gained 0.11 (0.04–0.20); within a phase the gain was −0.02 (−0.13 to +0.11) (Figure 3A). Within a phase, images alone reached 0.68 (0.58–0.77) and the timeline 0.64 (0.51–0.78), so within a phase the images gave about as much evidence as the timeline.

At UCSF the progression calls rest largely on enhancement. Enhancing tumour volume alone,

the simplest baseline, gave an AUC of 0.88 (0.79–0.95), above the locked image models; UCSF rated enhancement as increased at 27 of the 29 progression scans, and progression was dated a median of 7 days after the evaluated scan (Figure 3B). Time since diagnosis, which separates the classes at MU, gave 0.54 at UCSF. Among the 37 scans rated as showing increased enhancement, the single-scan image model reached 0.65 (0.44–0.84), so beyond that rating the measurements added little. At UCSF the AUCs therefore largely measure agreement with a progression call made from enhancement at the same scan, which is what PRISM's image pathway measures. The combined score's higher AUC at UCSF (Figure 2) rests partly on the interval between the two scans, which was shorter before progression.

Adding inputs to the image model raised its AUC on MU, and comparing within phases of care shows what each addition contributed (Figure 3C). Measurements of the current scan alone scored lowest, and adding change from earlier scans gave the image pathway of Section 3.1. Adding the interval since the previous scan, a flag for whether there was one and the smallest volume counted from the end of radiotherapy raised the AUC to 0.81, close to the timeline; within a phase these three inputs added nothing (−0.01, −0.06 to +0.04), because they encode the scan schedule. They form the imaging block of the planned model, so PRISM's image pathway leaves them out.

Deep networks fine-tuned to score UCSF scans directly learned above a permuted-label control (Table 2), but their scores tracked enhancing volume (correlation about 0.8) and their AUC stayed below that of enhancing volume alone; on MU they did not meet the planned gain over the timeline. Because the evidence behind a score differs by cohort and by situation, PRISM also states how far each score can be trusted.

3.4. The conformal sets meet their coverage threshold at all three centres

At the 80% level, the conformal sets of the combined score covered between 0.78 and 0.87 of scans in the three cohorts, meeting the threshold of 0.78 fixed in advance, a tolerance of 0.02 below the nominal level (Figure 4A). At UCSF the point estimate sat at the threshold and its interval reached below it (0.68–0.87). Most sets were informative, holding a single outcome for 62% to 75% of scans. Coverage fell short for the image pathway alone at LUMIERE and the timeline alone at UCSF, and it was uneven across situations within each cohort (Supplementary Section [S24]).

On MU, the warning that the two pathways disagree picked out the less accurate scans. The pathways disagreed on 40% of MU scans; the combined score's AUC was 0.90 where they agreed and 0.59 where they disagreed (difference 0.31, 0.18–0.43), and setting the disagreeing scans aside gained more than removing the same number matched for phase of care (Table 2; Figure 4B). At LUMIERE and UCSF disagreement did not pick out the less accurate scans. A stricter rule chosen after these results, flagging a scan only when the pathways disagree and its set holds both outcomes, is described in Supplementary Section [S24] (exploratory, not tested on new data).

The reliability card states, for each scan, how much measured evidence lies behind its situation, and for most scans in our cohorts this is little (exploratory). Across the three cohorts only four situations held at least 10 progression and 10 non-progression scans, one at MU and three at LUMIERE; every situation at UCSF had few cases (Figure 4C). At MU no situation in the first 12 weeks after radiotherapy held more than three progression scans. At LUMIERE scans in tested situations had lower error than scans with few cases; at MU the order was reversed, because the few-case scans there were mostly non-progression scans early after radiotherapy, which the models score easily

(Table 2). The clinician reads these statements in the report, which we evaluated next.

3.5. The reports are faithful to the record

Every evaluation report written by the local model passed the deterministic check against its record, so every number, flag and confidence statement in it could be traced to the record (114, 162 and 130 reports at MU, LUMIERE and UCSF; Figure 5). These counts come from a second evaluation, after one revision of the prompt, on patients not used in development. The first evaluation had failed at LUMIERE (0 of 178) because one sentence quoted a number that was in no cited field; we corrected the prompt and left the checker unchanged. The reports stated the evidence tier for all 220 scans in situations with few or no cases, and followed a changed field in 28 of 30 perturbed records; both misses were changes in enhancing volume from a previous value of zero. The clinician therefore reads the segmentation, both pathway scores and the confidence as they stand in the record. Of the seven open-weight models we compared, the smallest that passed in every cohort had 27 billion parameters and ran on one GPU (Supplementary Section [S25, second report evaluation, to be added]).

4. Discussion

PRISM gives the clinician separate image and treatment-timeline evidence at each scan, transfers to two external centres without refitting, and states how far its scores can be trusted there. At the development centre the timeline provided most of the evidence, largely because of when patients were scanned, and within a phase of care the images provided about as much. At UCSF, where few treatment dates are recorded, the image measurements provided the score, and it matched the enhancement-based call made at the same scan. Training at UCSF raised the image pathway further, so the locked results show transfer rather than the best a centre can reach. Because the pathways were kept apart, these differences could be measured at each centre, and the confidence and the card bring them into each report.

Because its modules are separate, PRISM runs where data are incomplete. The image pathway needs only the scans and their masks, which is what most public surveillance datasets hold; at LUMIERE and UCSF it ran without treatment dates. Where dates exist, the timeline is added as a second pathway, and the combined score, one ridge model on both sets of inputs, is shown beside both without changing the image model. A new cohort adds rows to the reliability card, so the tiers change as evidence accumulates and no confidence model has to be refitted.

Consider, as an illustration, a scan eight weeks after radiotherapy in which the enhancing tumour has grown. At MU, 1 of the 9 scans in this situation on the card was progression (exploratory). The timeline pathway scores such a scan low, because most scans at this point in treatment are not progression, and the image pathway scores it high, because enhancement has grown. PRISM's report would show both scores, state that they disagree, give a conformal set that holds both outcomes and record that the card has few cases for this situation. The clinician then sees that the system has no measured basis for a call at this scan, in the weeks when pseudoprogression is most common, and decides with information PRISM does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each of PRISM's components has been described before, and we are not aware of a system that

combines them. Automatic segmentation of post-treatment glioma with measurement of change between scans is established (Rudie et al. 2022), conformal prediction has been applied to glioma segmentation (Elyassirad et al. 2025), clinical and imaging data have been fused to separate pseudo-progression from progression (Gomaa et al. 2025), and model cards report performance by subgroup (Mitchell et al. 2019). The closest system we found combines treatment variables extracted by a language model with tumour volumes to assign a structured follow-up category at one centre, without an uncertainty attached to the output (Jabal et al. 2026). On MU, a radiomics model reached an AUC of 0.80 with a label assigned at the patient level and one held-out set (Christodoulou et al. 2026), so its result and ours answer different questions. On LUMIERE, a classifier of four response categories reached a mean AUC of 0.81 [cite Tikhonov 2025, arXiv:2509.06511; full reference not yet verified], and Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres; both address other labels.

This study has limits. PRISM was developed at one centre, with 126 grade 4 patients, and the two external tests are its only evidence of transfer. Progression is the progression documented by each centre, without expert re-reading of the scans; at UCSF it agrees closely with an enhancement call at the same scan, and a quarter of the non-progression scans there show no enhancement. Subgroups are small; UCSF contributes 29 progression scans, and its intervals are 0.15 to 0.25 wide. The coverage threshold was met on point estimates, and at UCSF the interval reached below it; the disagreement warning picked out less accurate scans only at MU. The three-step image comparison, the agreement rule, the card and the stricter flag rule are exploratory and need testing on new data. The report checker is lexical, so it can confirm that every number and statement is in the record but cannot judge how a clinician reads the report, and no clinician has yet read the reports. PRISM forecast progression within 120 days less well than it detected it at the scan, so we present it as a detection system and report the forecast as a secondary task.

The next step is to pool the rare situations across cohorts and train on them. Early progression after radiotherapy and growing enhancement without progression are rare at any one centre. The Burdenko and UCSD-PTGBM datasets add radiotherapy detail and advanced sequences, and each new cohort adds cases to the card's thin situations, which is where models fine-tuned on these long-tail cases could be tested. A reader study in which clinicians use PRISM's reports at the scan would show whether separate evidence and a stated confidence change their decisions.

Conclusion

PRISM shows that a surveillance system can keep its kinds of evidence apart, transfer them to other centres without refitting and state how far each has been tested there. In our three cohorts the images and the treatment timeline each contributed, in different proportions at each centre, and PRISM shows the clinician both scores and the evidence behind them at every scan.

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Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other open-weight models in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no patient-level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

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Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of de-identified datasets (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test; their commits are listed in Table 2. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

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Table 1. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development (repeated cross-validation)	External, locked	External, locked
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 index scans (median 2)	Up to 10 index scans (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	None; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Progression documented as	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62.5 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: progression / non-progression scans (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; non-progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary 120-day task at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG. Age is at diagnosis at MU-Glioma-Post and at surgery at LUMIERE.

[b] Positives are the scan at which progression was documented; negatives are defined per cohort in Section 2.2. The LUMIERE detection set has more patients than the 120-day cohort because it includes patients who contribute only a progression scan. At MU-Glioma-Post, 47 progression scans of patients with no earlier non-progression scan are excluded.

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six-week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 2. Questions fixed in advance and their answers. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Result	Met
Detection on MU-Glioma-Post			
Do image measurements add to the timeline?	Planned model ≥ 0.75 (lower bound ≥ 0.65) and gain ≥ 0.03 with CI above 0	Model 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)	No
Do fine-tuned deep networks add?	Gain ≥ 0.03 with CI above 0; permuted-label control covers 0.5	No network added; control 0.483 (0.436–0.527)	No (control valid)
Does the locked model transport to LUMIERE?	External AUC ≥ 0.65	Lower bound above 0.65 (Section 3.2)	Yes
UCSF-ALPTDG			
Q1. Does the image pathway transport?	AUC ≥ 0.65, lower bound > 0.5	Lower bound 0.603	Yes
Q2. Do images add to clinical data without a timeline?	Gain with CI above 0	+0.248 (+0.096 to +0.392)	Yes
Q3. Does training at UCSF-ALPTDG do better than the locked model?	Descriptive	Higher when trained there (Section 3.2)	Descriptive
Q4. Do deep networks learn at UCSF-ALPTDG?	Above the permuted-label control; control covers 0.5	+0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)	Yes
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)			
S1. Do conformal sets keep their coverage?	Coverage ≥ 0.78 at the 80% level in each cohort	Combined score met in all three (Section 3.4); images at LUMIERE 0.708 and timeline at UCSF-ALPTDG 0.696	Yes (combined)
S2. Do flags pick out less accurate scans?	AUC unflagged minus flagged > 0, CI above 0	Disagreement met at MU-Glioma-Post (Section 3.4); LUMIERE +0.105 (−0.049 to +0.266); UCSF-ALPTDG −0.019 (−0.218 to +0.175); any flag at MU-Glioma-Post −0.170 (−0.503 to +0.147)	MU only
S3. Does setting flagged scans aside beat matched removal?	Retained AUC minus matched removal > 0, CI above 0	Disagreement met at MU-Glioma-Post (Section 3.4); LUMIERE +0.051 (−0.028 to +0.144); UCSF-ALPTDG −0.004; set holding both outcomes met at UCSF-ALPTDG, +0.073 (+0.014 to +0.138)	MU only (disagreement)
S4. Do the card's tiers mean something?	Descriptive: tested below few cases in Brier score and error	Brier 0.187 vs 0.258 at LUMIERE; reversed at MU-Glioma-Post, 0.222 vs 0.147	Descriptive
S5. Are reports faithful to the record?	≥ 95% pass the checker in each cohort	First evaluation: LUMIERE 0 of 178; second evaluation: all passed (Section 3.5)	No, then yes
S6. Is the tier stated where evidence is thin?	100%	All (Section 3.5)	Yes
S7. Do reports follow	≥ 27 of 30 perturbed	29 of 30 in the first evaluation;	Yes

Figure legends

[PLACEHOLDER: fig1_system.pdf]

PRISM is built of modules, each answering a known challenge: scan and record; segmentation and measurements; the timeline pathway; the image pathway; the combined score; the confidence (conformal set, agreement, card); the local report. One worked example scan.

Figure 1. PRISM is built of modules, each answering a known challenge. A scan and its record enter two pathways: the timeline pathway (clinical data and the patient's position in treatment) and the image pathway (measurements from the segmentation masks supplied with each cohort). The combined score, one ridge model on both sets of inputs, is shown beside them. The confidence has three parts: a conformal prediction set at the 80% level, a statement of whether the two pathways agree, and the reliability card's tier for the scan's situation. A local language model writes the report from the record, and a deterministic checker tests every sentence against it. One example scan is followed through each module.

[PLACEHOLDER: fig2_transport.pdf]

AUC by pathway (clinical alone, timeline, images, combined) and cohort (MU internal; LUMIERE and UCSF locked; UCSF-trained as a reference).

Figure 2. The models transfer to two external centres without refitting. AUC for detecting the scan at which progression was documented, by pathway (clinical data alone, timeline, images, combined) and cohort: MU-Glioma-Post (repeated cross-validation, intended use); LUMIERE and UCSF-ALPTDG (models fitted once on MU-Glioma-Post and applied unchanged); and UCSF-ALPTDG with models trained there, as a reference. At LUMIERE and UCSF-ALPTDG the timeline is the reduced version, and the clinical-only model at LUMIERE is exploratory. At UCSF-ALPTDG the images are shown for the single-scan model and for the image pathway with change features, which is the system's image pathway; the combined score there includes the interval between the two scans. Bars are 95% patient-bootstrap CIs.

[PLACEHOLDER: fig3_evidence.pdf]

(A) MU: the timeline's gain lies between phases of care. (B) UCSF: enhancing volume alone scores above the image models. (C) The image-model ladder (exploratory).

Figure 3. What each score rests on depends on the cohort. (A) MU-Glioma-Post: the gain from adding the timeline to clinical data and images, across phases of care and within them. (B) UCSF-ALPTDG: AUC of enhancing tumour volume alone beside the locked image models, and the share of progression and non-progression scans that UCSF rated as showing increased enhancement. (C) The image model on MU-Glioma-Post built in three steps (current scan; with change from earlier scans; with schedule inputs), across and within phases of care. All panels are exploratory. Bars are 95% patient-bootstrap CIs.

[PLACEHOLDER: fig4_confidence.pdf]

(A) Conformal coverage at the 80% level by cohort and pathway. (B) AUC with and without the disagreement warning. (C) The reliability card: situations by cohort with their evidence tier.

Figure 4. The conformal sets meet their coverage threshold, and at MU-Glioma-Post disagreement marks the less accurate scans. (A) Coverage of the 80% conformal sets by cohort and pathway; the dashed line is the threshold fixed in advance, met by the combined score on point estimates in all three cohorts. (B) AUC of the combined score where the pathways agree and where they disagree, and after the disagreeing scans are set aside, compared with removal matched for phase of care, at MU-Glioma-Post; at LUMIERE and UCSF-ALPTDG disagreement did not pick out the less accurate scans. (C) The reliability card: situations (phase of care by change in enhancing tumour) in each cohort, shaded by tier (tested, few cases, not tested). The agreement rule and the card are exploratory.

[PLACEHOLDER: fig5_report.pdf]

One rendered report (segmentation figure, pathways, combined score, confidence) and faithfulness by cohort.

Figure 5. The report states the evidence, faithful to the record. Left: one report as the clinician receives it, with the segmentation figure and its volumes, the two pathway scores, the combined score, the conformal set, the agreement statement and the card's tier. Right: share of reports passing the deterministic check, by cohort, in the second evaluation, after one revision of the prompt following a failure at LUMIERE (Section 3.5).